

solid-state technology in non-flexible applications. A prime example is Prologium, indicating that advancements in solid-state technology for EVs could also enhance the flexible battery sector.

The flexible battery market remains fragmented, offering opportunities for new entrants with innovative technologies to establish niches, unlike the highly consolidated and resource-intensive EV and ESS sectors. Over the next decade, the flexible battery market may see more consolidation, with the emergence of one or two major players per battery chemistry. However, rapid consolidation is not expected due to the slow pace of development and the market's moderate size.

Additionally, there is potential for larger firms that previously exited the flexible battery space to re-enter with technologies designed for specific applications, such as smartphones. These re-entries may remain niche but could have a significant impact on the broader battery market.

Powering wearable tech

Wearable technology, especially augmented reality (AR), continues to expand rapidly, already finding use in industrial applications and moving towards broader commercial use. If companies in this sector start incorporating flexible batteries that meet necessary specifications, the market could experience significant growth.

Despite the excitement surrounding flexible batteries in consumer electronics, their practical use in smartphones remains limited due to the requirement for all components to be flexible—a costly and challenging requirement. Foldable phones, which offer adaptability without requiring full flexibility, already meet much of this demand.

The integration of flexible batteries in wearables improves user comfort and supports diverse form factors, from smart rings to wristwatches. This

Success factors in the flexible battery market

In an increasingly competitive landscape, companies in the flexible battery market are finding that the key to success lies in innovation and integration. Understanding the dynamics between manufacturing high-quality products and effectively incorporating them into consumer-ready solutions is essential.

- **Integrated product development.** The most successful players in the flexible battery market are those that not only produce high-quality batteries but also seamlessly integrate these batteries into complete products. Historically, companies that maintain control over the entire product development process have experienced greater success than those that outsource integration.
- **Collaborative partnerships.** Forming partnerships with OEMs has proven advantageous, especially for creating specialised applications like smart labels, smart cards, and medical sensors. These collaborations cater to market demands for integrated solutions and contribute significantly to a company's success.
- **Specialisation case study.** A prime example of market adaptation is a zinc-carbon battery manufacturer that shifted to develop temperature-tracking sensors for medical use during the Covid-19 pandemic. This strategic move to address an urgent medical need increased the company's market presence and underscored the value of targeted product development for industry leadership.

By focusing on strategic integration and specialised partnerships, companies can leverage their expertise in flexible battery technology to meet specific industrial needs, enhancing their offerings to OEMs and boosting overall market adoption and customer satisfaction.

makes wearables an ideal market for the growth of flexible batteries.

Smart labels and RFID tags also provide substantial growth opportunities, requiring flexibility to adapt to diverse and changing surfaces. This need underscores the potential for flexible batteries in industrial applications, where traditional rigid batteries fall short.

Currently, flexible battery production is mostly limited to pilot lines and small-scale setups. Scaling up production to meet rising demand poses a significant challenge but also a substantial opportunity. Standardising battery shapes and sizes across the industry could reduce manufacturing costs, simplify production processes, and lessen the need for customised equipment.

The future growth of flexible batteries is poised to concentrate in two distinct areas, reflecting the major categories of these technologies. For thin flexible batteries, significant expansion is anticipated in the logistics sector, including smart labels and RFID tags, driven by focused efforts of established companies and

increasing investments linked to the rising interest in IoT applications. Conversely, bulkier battery types, such as solid-state and advanced lithium-ion batteries, are expected to find substantial growth opportunities within the wearable technology market, particularly in medical wearables. Despite stringent healthcare regulations, the increasing number of trials and testing contracts with OEMs in the medical sector signals robust growth over the next decade. While wearables will likely see considerable expansion across various sub-sectors, with wrist-worn devices leading, the growth of e-textiles may remain subdued due to its smaller market size, positioning smart labels, RFID tags, and medical wearables as the primary application markets driving the adoption of flexible batteries in the coming years. **EFY**

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